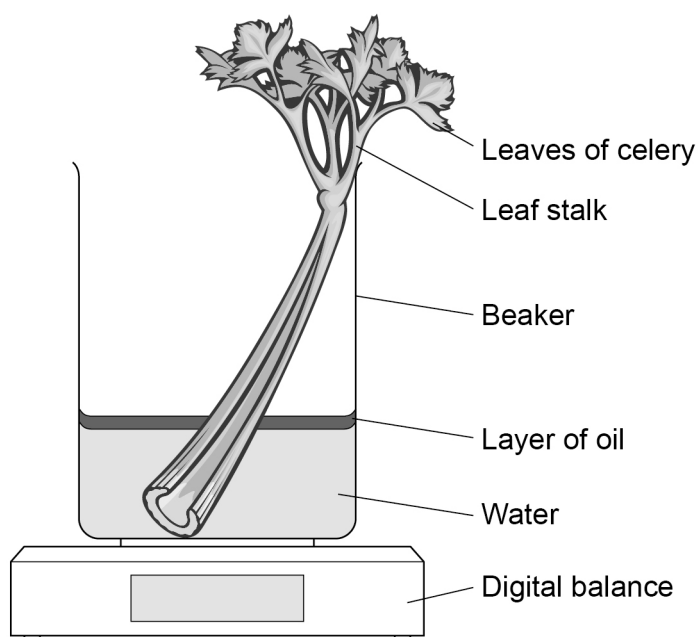


0 5

A student used the apparatus shown in **Figure 5** and a digital balance to determine the rate of water movement in a celery stalk in grams per hour per group of xylem vessels.

Figure 5



0 5 . 1

The student measured the time taken for water movement.
Give **two** other measurements he made to calculate the rate of water movement.

[2 marks]

1 _____

2 _____

0 5 . 2

Give the reason for adding a layer of oil to the water in the beaker.

[1 mark]

Question 5 continues on the next page

Turn over ►



0	5	.	3
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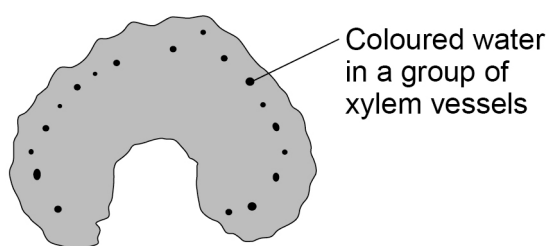
A different student used coloured water to investigate the movement of water in leaf stalks of celery.

During the procedure she:

- cut equal lengths of stalk from each plant
- put the cut end of each stalk into coloured water
- left these stalks to take up the coloured water for 20 minutes
- used a sharp scalpel to cut slices from the stalks at 1 mm intervals until she reached a slice with no coloured water.

Figure 6 shows a slice of leaf stalk with coloured water inside groups of xylem vessels.

Figure 6



Explain why coloured water moved up the stalks.

[3 marks]

[illegible]

0 5 . 4

The student used a sharp scalpel to cut the celery. Describe how she should ensure she handled the scalpel safely during this procedure.

[2 marks]

The student measured the distance the coloured water had travelled in eight celery stalks.
Her results are shown in **Table 1**.

Table 1

Distance / mm							
70	35	40	35	30	80	42	44

0 5 . 5

The student had to choose whether to summarise her measurements by calculating the mean, the median or the mode.

Circle the most appropriate measure for this set of measurements.
Give a reason for your choice and find the value using the measurements from all eight stalks.

[2 marks]

Mean* Median* Mode*

*circle one word.

Reason:

Calculation:

Answer = _____

10

Turn over ►